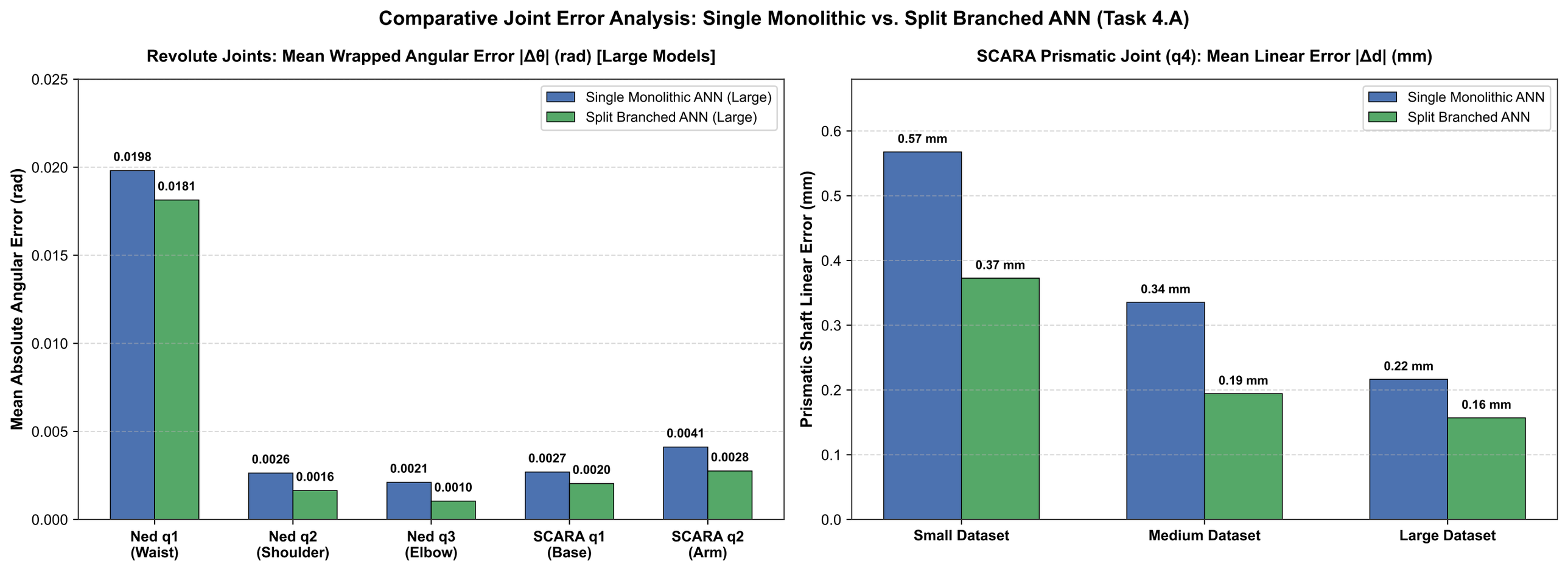
Comparative Joint Error Analysis: Single Monolithic vs. Split Branched ANN (Task 4.A)
Revolute Joints: Mean Wrapped Angular Error |Δθ| (rad) [Large Models]
Mean Absolute Angular Error (rad)
0.025
0.020
0.015
0.010
0.005
0.000
Single Monolithic ANN (Large)
Split Branched ANN (Large)
0.0198
0.0181
0.0026
0.0016
0.0021
0.0010
0.0027
0.0020
0.0041
0.0028
Ned q1 (Waist)
Ned q2 (Shoulder)
Ned q3 (Elbow)
SCARA q1 (Base)
SCARA q2 (Arm)
SCARA Prismatic Joint (q4): Mean Linear Error |Δd| (mm)
Prismatic Shaft Linear Error (mm)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Single Monolithic ANN
Split Branched ANN
0.57 mm
0.37 mm
0.34 mm
0.19 mm
0.22 mm
0.16 mm
Small Dataset
Medium Dataset
Large Dataset